# Review for First Semester Exam Grade 9

## A. Multiple choice questions

**1.** Which of the following is the formula of moment?

**a.** $F = M \times d$ **b.** $M = \dfrac{F}{d}$

**c.** $M = Fd$ **d.** $d = M \times F$

**2.** Which of the following is the mechanical advantage of lever?

**a.** $MA = \dfrac{F_R}{F_E} = \dfrac{d_E}{d_R}$ **b.** $MA = \dfrac{F_R}{F_E} = \dfrac{L}{h}$

**c.** $MA = \dfrac{F_R}{F_E} = \dfrac{R_R}{r_E}$ **d.** $MA = \dfrac{F_R}{F_E} = \dfrac{R}{r}$

**3.** The lever which has the effort in between the load and the fulcrum is the:

**a.** First class **b.** Second class
**c.** Third class **d.** None of these

**4.** A wire which is complete the circuit by providing a return path to the supply for current called:

**a.** Live wire **b.** Neutral wire
**c.** Earth wire **d.** None of these

**5.** Which of the following is electrical device using for breaking circuit?

**a.** Switch **b.** Fuse
**c.** Circuit breaker **d.** None of these

**6.** How many types of the electric current?

**a.** Two **b.** Three
**c.** Four **d.** Five

**7.** The electric current which is flowing back and forth approximately 50 times a second is known as

**a.** Specific heat capacity **b.** Electric charges
**c.** Frequency **d.** Potential difference

**8.** How many main types of pulley system?

**a.** Two **b.** Three
**c.** Four **d.** Five

**9.** Which statement is not TRUE for specific heat capacity?

**a.** specific heat capacity is the amount of heat energy that 1 g of substance needed to change its temperature by 1 degree Celsius.

**b.** specific heat capacity is the amount of heat energy that 1kg of substance needed to change its temperature by 1 degree Celsius.

**c.** specific heat capacity is the amount of heat energy that 1 kg of substance needed to change its temperature by 1 K.

**d.** specific heat capacity is measured in $J/kg^0C$.

**10.** Which of the following is the energy transfer?

**a.** $E = Pt$ **b.** $E = VI$

**c.** $E = \dfrac{V^2}{R}$ **d.** $E = RI^2$

**11.** Which of the following is not an electrical power?

**a.** $P = RI$ **b.** $P = VI$

**c.** $P = \dfrac{V^2}{R}$ **d.** $P = RI^2$

**12.** Which statement is NOT true for the circuit breaker?

**a.** breaks the circuit when the current over flow
**b.** prevents from catching fire in the house
**c.** melting when the current goes above the certain value
**d.** use electromagnetic effect to break the circuit

**13.** A resistance light bulb is connected to 12 V battery. If the power of the bulb is 24W, what will be the resistance of the bulb?

**a.** 2 ohms **b.** 0.5 ohms

**c.** 6 ohms **d.** 48 ohms

**14.** A 60-tooth gear is connected to another gear. When the 60-tooth gear completed 4 revolutions, the second gear completed 6 revolutions. What will be the number of teeth of second gear?

**a.** 40 **b.** 90

**c.** 1440 **d.** 20

**15.** All the tools below are the first-class lever except:

**a.** scissors **b.** staplers
**c.** pliers **d.** see-saw

**16.** When an object is heated, its temperature is increased upon on the factors below except:

**a.** the amount of energy that the object absorbed
**b.** the mass of the object
**c.** the size of the object
**d.** the substance of the object is made

**17.** Two downward forces, $F$ and 30N, act on a light uniform plank of length $L$ at the two ends as shown. Calculate the force $F$ to make the plank balance.




**a.** 10N **b.** 30N

**c.** 40N **d.** 120N

**18**. Which of the following is an example of a first-class lever?
**a.** Nutcracker **b.** See-saw
**c.** Fishing rod **d.** Wheelbarrow
**19**. What is the mechanical advantage of a lever?
**a.** The ratio of output force to input force
**b.** The distance moved by the load
**c.** The efficiency of the lever
**d.** The angle of the lever's arm
**20**. In a third-class lever, where is the effort located?
**a.** Between the load and the fulcrum
**b.** At the opposite end of the fulcrum
**c.** Next to the load
**d.** Equally distributed
**21.** What is the primary purpose of an inclined plane?
**a.** To lift heavy objects using less force
**b.** To reduce friction between objects
**c.** To increase the velocity of an object
**d.** To convert rotational motion into linear motion
**22.** What happens to the mechanical advantage of an inclined plane as its slope becomes steeper?
**a.** It increases **b.** It decreases
**c.** It remains the same **d.** It becomes infinite
**23.** Which type of pulley provides a mechanical advantage?
**a.** Fixed pulley **b.** Movable pulley
**c.** Simple pulley **d.** All types of pulleys
**24.** What is the main purpose of a pulley system?
**a.** To increase the speed of motion
**b.** To change the direction of a force or reduce the effort needed
**c.** To decrease the size of an object
**d.** To improve rotational energy
**25.** In a gear system, what determines the speed ratio between two gears?
**a.** The material of the gears
**b.** The diameter or number of teeth on each gear
**c.** The distance between the gears
**d.** The color of the gears
**26.** What is the primary function of a belt in a belt and pulley system?
**a.** To transmit rotational motion from one shaft to another
**b.** To increase mechanical advantage
**c.** To convert linear motion into rotational motion
**d.** To reduce friction
**27.** Combined effect of several forces is known as
**a.** gravity **b.** resultant force
**c.** normal force **d.** weight
**28.** An electric heater is rated at $500\ \mathrm{W}$. How much energy is delivered to the heater in $\mathrm{half\ an\ hour}$?
**a.** $1000\ \mathrm{J}$ **b.** $250\ \mathrm{J}$
**c.** $900\ \mathrm{kJ}$ **d.** $900\ \mathrm{W}$

**29.** Which of the following is an example of a wheel and axle?
**a.** Door handle **b.** Screw
**c.** Pulley **d.** Inclined plane
**30.** A lever is used to lift a rock weighing $100\ \mathrm{N}$. If the distance from the fulcrum to the effort is $2\ \mathrm{m}$ and the distance from the fulcrum to the load is $0.5\ \mathrm{m}$, what is the effort required?
**a.** $25\ \mathrm{N}$ **b.** $50\ \mathrm{N}$ **c.** $75\ \mathrm{N}$ **d.** $100\ \mathrm{N}$
**31.** A crowbar is used to lift a heavy object. The fulcrum is placed $30\ \mathrm{cm}$ from the load and $90\ \mathrm{cm}$ from the effort. What is the mechanical advantage of the lever?
**a.** $0.33$ **b.** $1$ **c.** $3$ **d.** $6$
**32.** A $200\ \mathrm{N}$ box is pushed up an inclined plane that is $4\ \mathrm{m}$ long and $1\ \mathrm{m}$ high. What is the mechanical advantage of the inclined plane?
**a.** $1$ **b.** $2$ **c.** $4$ **d.** $8$
**33.** A worker pushes a crate weighing $500\ \mathrm{N}$ up a ramp with a force of $200\ \mathrm{N}$. If the ramp is $5\ \mathrm{m}$ long and $2\ \mathrm{m}$ high, what is the efficiency of the inclined plane?
**a.** $50\%$ **b.** $75\%$ **c.** $80\%$ **d.** $100\%$
**34.** A pulley system with 3 supporting ropes is used to lift a $300\ \mathrm{N}$ load. What force is needed to lift the load (ignoring friction)?
**a.** $50\ \mathrm{N}$ **b.** $100\ \mathrm{N}$ **c.** $150\ \mathrm{N}$ **d.** $300\ \mathrm{N}$
**35.** A person uses a movable pulley to lift a $400\ \mathrm{N}$ object. If the person applies a force of $200\ \mathrm{N}$, what is the mechanical advantage of the pulley system?
**a.** $1$ **b.** $2$ **c.** $4$ **d.** $8$
**36.** Two gears are connected. Gear A has $20\ \mathrm{teeth}$ and rotates at $120\ \mathrm{rpm}$. Gear B has $40\ \mathrm{teeth}$. What is the rotational speed of Gear B?
**a.** $30\ \mathrm{rpm}$ **b.** $60\ \mathrm{rpm}$ **c.** $120\ \mathrm{rpm}$ **d.** $240\ \mathrm{rpm}$
**37.** A wheel and axle system is used to lift a bucket weighing $150\ \mathrm{N}$. The radius of the wheel is $0.5\ \mathrm{m}$, and the radius of the axle is $0.1\ \mathrm{m}$. What is the mechanical advantage of the system?
**a.** $2$ **b.** $3$ **c.** $5$ **d.** $10$
**38.** A wheel and axle system has a mechanical advantage of $4$. If a force of $50\ \mathrm{N}$ is applied to the wheel, what is the maximum weight that can be lifted by the axle?
**a.** $50\ \mathrm{N}$ **b.** $100\ \mathrm{N}$ **c.** $150\ \mathrm{N}$ **d.** $200\ \mathrm{N}$

**39**. How many types of wire are commonly used in AC
**a.** Two **b.** Three **c.** Four **d.** Five
**40**. Which wire carries the current to the appliance?
**a.** Neutral wire **b.** Earth wire
**c.** Live wire **d.** Fuse wire
**41**. What is the main purpose of the neutral wire?
**a.** To prevent electric shocks
**b.** To complete the circuit by returning path
**c.** To protect the appliance
**d.** To carry excess current to the ground
**42.** Which of the following wires is used for safety and prevents electric shocks?
**a.** Live wire **b.** Neutral wire
**c.** Earth wire **d.** None of the above
**43.** What color is commonly used to identify the live wire in most wiring systems?
**a.** Blue **b.** Brown
**c.** Green/Yellow **d.** Black
**44.** Why is the earth wire necessary in appliances with metal casings?
**a.** To enhance the efficiency of the appliance
**b.** To protect the wiring inside the appliance
**c.** To ensure excess current is diverted safely to the ground
**d.** To supply extra current
**45.** Which of the following is the potential difference between the live wire and the neutral wire in a standard household supply?
**a.** 0 V **b.** 230 V **c.** 400 V **d.** 50V
**46.** What is the primary purpose of the third pin in a 3-pin plug?
**a.** To complete the circuit
**b.** To provide an earthing connection for safety
**c.** To control the voltage
**d.** To reduce power consumption
**47.** Why do 2-pin plugs not have an earth pin?
**a.** They are designed for devices that use low voltage
**b.** The appliances have double insulation for safety
**c.** The earth pin is unnecessary for any device
**d.** The current is too high
**48.** Which type of appliances commonly use 2-pin plugs?
**a.** Appliances with metallic casings
**b.** Appliances that are double insulated
**c.** Appliances requiring high power
**d.** Appliances used in industrial setups
**49.** Specific heat capacity of aluminum when 26400 J of heat energy is supplied to 2 kg block and it's temperature rises from 20 °C to 35 °C is
**a.** 1000 J/kg.K **b.** 70 J/kg.K
**c.** 400 J/kg.K **d.** 880 J/kg.K
**50.** What does "double insulation" mean in electrical appliances?
**a.** The appliance has an extra layer of wiring
**b.** Both the live and neutral wires are insulated
**c.** The appliance has no metallic casing and is protected by an additional layer of insulation
**d.** The appliance uses two earth wires
**51.** Which of the following is true for a 3-pin plug?
**a.** It provides only live and neutral connections
**b.** It is used for appliances with double insulation
**c.** It ensures safety by grounding the appliance through the earth pin
**d.** It works only on high-voltage appliances
**52.** Why are metallic appliances without double insulation required to use a 3-pin plug?
**a.** To reduce energy loss
**b.** To enhance appliance performance
**c.** To provide a safety connection to the ground in case of a fault
**d.** To increase the lifespan of the appliance
**53.** What is the primary purpose of a two-way switch?
**a.** To control an appliance from two different locations
**b.** To regulate the voltage supplied to a circuit
**c.** To provide safety against electric shocks
**d.** To reduce power consumption
**54.** In a two-way switch setup, what connects the two switches to allow control from both points?
**a.** Earth wire **b.** Live wire
**c.** Neutral wire **d.** A common wire (loop wire)
**55.** Which of the following is a common application of a two-way switch?
**a.** To operate a fan at variable speeds
**b.** To control lights in a staircase from the top and bottom
**c.** To provide earthing for appliances
**d.** To protect against electric shocks
**56.** Which of these scenarios can lead to an electric shock?
**a.** Touching an insulated live wire
**b.** Touching a live wire while standing on a wet surface
**c.** Handling a properly earthed appliance
**d.** Using a two-way switch correctly
**57.** Which of the following reduces the risk of electric shock in household wiring?
**a.** Using a two-way switch instead of a single-way switch
**b.** Ensuring all live wires are properly insulated
**c.** Using appliances with metallic casings
**d.** Avoiding the use of earthing connections
**58.** The point where the whole weight of an object acts is called
**a.** Central point **b.** center of gravity
**c.** edge **d.** center

**59.** No Earth wire is required for TV set because
**a.** The power consumed is less
**b.** TV does not have heating element
**c.** TV is made from plastic
**d.** The current is too small.

**60.** If a heater draws the 5 A current from a 220 V supply, what is the voltage on the neutral wire?

**a.** V= 0 V **b.** V = 5 V
**c.** V = 120 V **d.** V = 240 V

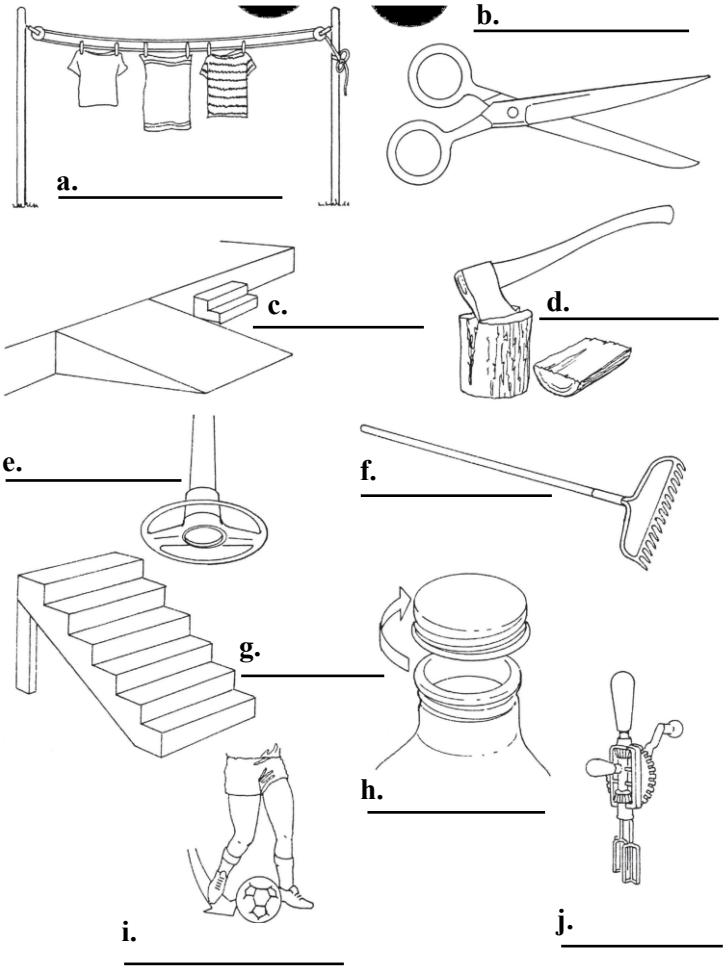
**61**. Which is the most appropriate fuse rating for a 240V, 1.1 kW heater?

**a.** $I = 1$ A **b.** $I = 5$ A
**c.** $I = 13$ A **d.** $I = 3$ A

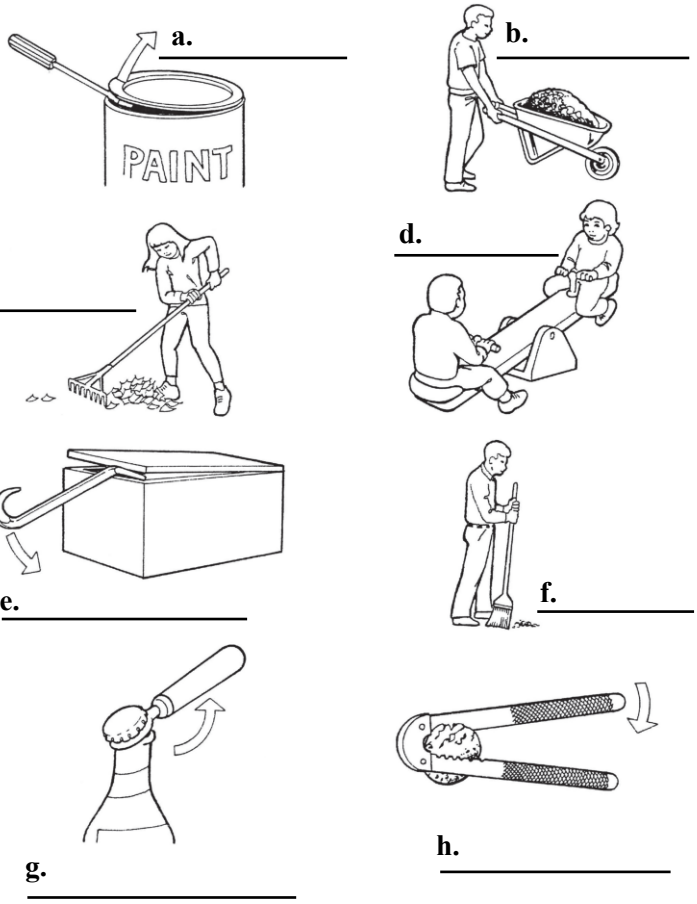
**62.** A seesaw is an example of a:

**a.** Wheel and axle **b.** Pulley
**c.** Lever **d.** screw

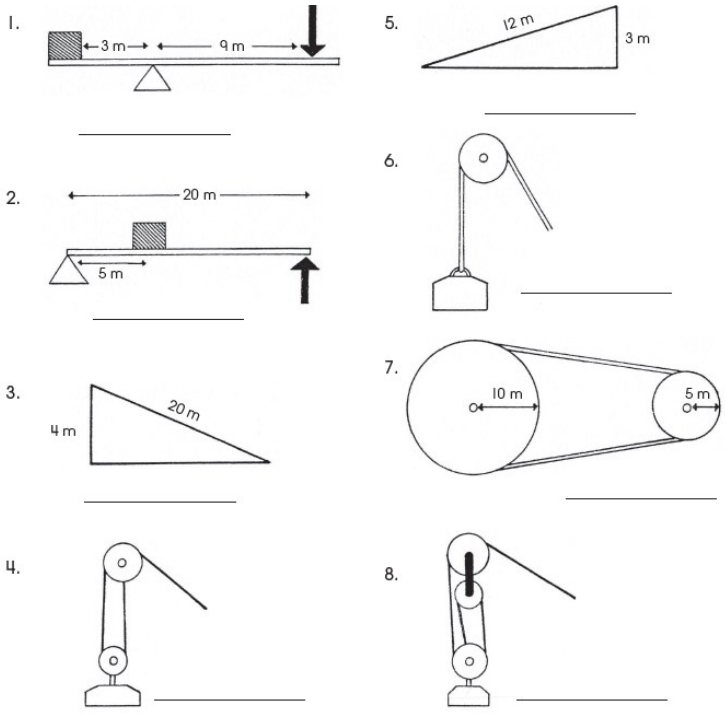
**63.** Identify the type of simple machine shown in each picture.




**64.** Classify each lever as first, second, or third class.




**65.** Identify the mechanical advantage of each simple machine.




**66.** How would you construct a compound pulley with an ideal mechanical advantage of 10.

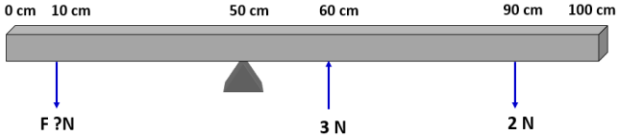
## B. Matching question

**67.** Match the word with the correct definition or corresponding phrase.

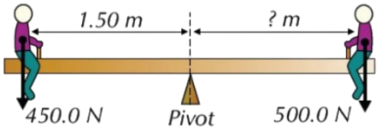
| | |
|---|---|
| **1.** ___ moment | A. is the ration of output force to input force. |
| **2.** ___ lever | B. is the thermal energy transfer between two substances. |
| **3.** ___ frequency | C. is the flowing of the electric charge in the circuit. |
| **4.** ___ pulley | D. is the wheel with teeth on the outer edges. |
| **5.** ___ efficiency of pulley | E. is loop of flexible material used to mechanically link two or more rotating shaft. |
| **6.** ___ electric current | F. is a simple machine used to changes the direction of a force, often to lift a load. |
| **7.** ___ inclined plane | G. is the electrical device used to turn on and off the appliances. |
| **8.** ___ Gear | H. is the following of electric current back and forth approximately 50 times a second. |
| **9.** ___ belt system | I. is the ratio of output work and input work. |
| **10.** __ heat energy | J. is the wire which is connected directly to high voltage and deliver electric current to appliances. |
| **11.** __ circuit breaker | |
| **12.** __ switch | K. is the rate at which the energy transfer or work done. |
| **13.** __ live wire | L. is the energy need to raise temperature of mass 1 kg by 1 degree Celsius. |
| **14.** __ mechanical advantage | M. is a rigid rod, pivoted about a fixed point or axis, which known as fulcrum. |
| **15.** __ specific heat capacity | N.is the flat supporting surface tiled at angle, with one end higher than the other. |
| **16.** __ power | O. is the product between force and perpendicular distance. |
| **17.** __ Center of gravity | P. is an electromagnetic device that breaks a circuit when the current goes above a certain value. |
| | Q: The point where the whole weight of an object acts or seem to acts. |

## C. Work problems

**68.** A 1 m long uniform light ruler is being balanced as shown. The force $F$ is suspended at 40 cm from the support, 3 N is suspended at 10 cm from the support, and 2 N is suspended at 40cm from the support. Calculate the unknown force $F$.




**69**. The diagram below shows two children on a seesaw. If the child on the left stays where they are, how far from the pivot should the child on right sit to balance the seesaw?




**70.** A worker pushes a box weighing 1500 N up a 6.0-meter-long ramp and places it on top of the plane at 1.50-meter height from the ground.

**a.** Calculate the mechanical advantage of using an inclined plane

**b.** Calculate the effort needed to push the box.

**71.** For the following simple see-saw calculate:

**a.** the load that could be lifted

**b.** the mechanical advantage (assume the lever is 100%efficient).

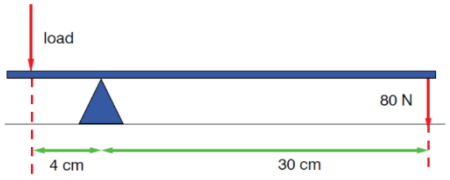



*Figure 5.44 A simple see-saw*

**72.** A television set having a power rating of 120 W and electric lawnmower of power rating 1 kW are both connected to a 240 V supply. If 3 A, 5 A and 10 A fuses are available state which is the most appropriate for each appliance.

**73.** A simple wheel and axle is used to lift a bucket of water out of a well. The radii of the wheel and axle are 20 cm and 4 cm, respectively. Determine:

**a)** the velocity ratio (and so the IMA)

**b)** the theoretical effort required to lift a load of 30 N assuming no energy losses

**c)** the efficiency if the actual effort required is 10 N.

**74.** An inclined plane rises to a height of 2 m over a distance of 6 m. Calculate:

**a)** the angle of the slope

**b)** the VR (and so IMA) of the inclined plane

**c)** the theoretical force required to push an object with a mass of 200 kg up the slope.

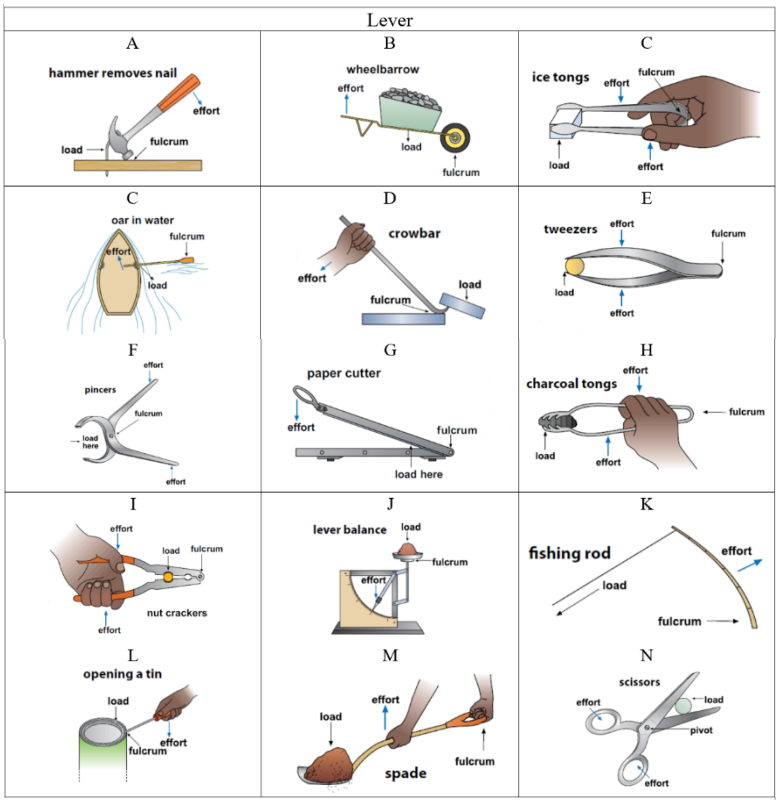
**75.** An electric kettle with $8.75\,\Omega$ of resistance is operating with $87.50\,\text{V}$ of voltage.

**a.** Find the electric current in this circuit

**b.** Calculate the electrical energy for using this electric kettle in 10 min.

**c.** If this electric kettle containing 10 kg of water, find the temperature change of the water. Assume that only 80% of electrical energy has absorbed by the water. Given: $c = 4200\,\text{J/kg.}^{\circ}\text{C}$

**76.** Identify whether each of the following is a first class, second class, or third-class lever.




**77. (i)** State the equation linking electrical power, current and voltage.

**(ii)** Determine the appropriate fuse rating for the following appliances. You can choose from a 3 A, 5 A or 13 A fuse.

**(a)** A 2.5 kW electric kettle operating on a 230 V supply.

**(b)** An 800 W electric drill operating on a 230 V supply.

**(c)** A 40 W computer operating on a 16 V supply.

**78.** An electric space heater draws 15.0 A from a 120-V source. It is operated, on the average, for 5.0 h each day.

**a.** How much power does the heater use?

**b.** How much energy in kWh does it consume in 30 days?

**c.** At \$0.12 per kWh, how much does it cost to operate the heater for 30 days?

**79.** A digital clock has a resistance of 12,000 Ω and is plugged into a 115-V outlet.

**a.** How much current does it draw?

**b.** How much power does it use?

**c.** If the owner of the clock pays \$0.12 per kWh, how much does it cost to operate the clock for 30 days?

**80.** A 120-V water heater takes 2.2 h to heat a given volume of water to a certain temperature. How long would a 240-V unit operating with the same current take to accomplish the same task?

**81.** A block of weight 5000 N is pushed up a slope by a force of 250 N. Assume there is no friction. Calculate:

**a)** the actual mechanical advantage

**b)** the velocity ratio

**c)** the length of the slope if the height of the slope is 10 m.

**82.** An inclined plane is 100 m long and at an angle of 20° to the horizontal. The AMA of the slope is two. Calculate:

**a)** the effort required to push a 7200 N block up the slope

**b)** the ideal mechanical advantage

**c)** the efficiency of the slope.

**83.** Use the information in Table 18.8 to work out the missing values **a**–**j**. Assume that electricity costs 16p per unit.

| Appliance | Power | Time | kWh/unit | Cost |
|---|---|---|---|---|
| Desktop computer | **a** | 3 hours | 1.3 | **b** |
| Microwave oven | 870 W | **c** | 0.87 | **d** |
| Television | **e** | 12 hours | 1.0 | **f** |
| Energy efficient lamp | 9 W | **g** | $4.5 \times 10^{-2}$ | **h** |
| Kettle | 2 kW | 3 minutes | **i** | **j** |

**84.** A 100-W light bulb is 20% efficient. That means 20% of the electric energy is converted to light energy.

**a.** How many joules does the light bulb convert into light each minute it is in operation?

**b**. How many joules of heat does the light bulb produce each minute?

**85.** The resistance of an electric stove element at operating temperature is 11 Ω.

**a.** If 220 V are applied across it, what is the current through the stove element?

**b.** How much energy does the element convert to thermal energy in 30.0 s?

**c.** The element is used to heat a kettle containing 1.20 kg of water. Assume that 65 percent of the heat is absorbed by the water. What is the water's increase in temperature during the 30.0 s?

**86. Fig. 6.30** shows a pulley system. An effort of 113 N is required to lift a load of 180 N.

**(a)** What distance does the effort move when the load moves 1 m?

**(b)** Find the work done by the effort.

**(c)** Find the work done on the load.



*Fig 6.30 A pulley system*